

INTERNSHIP IN ROBOTICS AND AUTOMATION

INTERNSHIP REPORT

*Submitted to the APJ Abdul Kalam Technological University
in partial fulfillment of requirements for the award of degree*

*Master of Technology
in
Robotics and Automation
by*

SHARANSUNDAR S (TVE24ECRA13)



INTERNSHIP

223IIA100

**DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
COLLEGE OF ENGINEERING TRIVANDRUM
KERALA
JULY 2025**

DEPT. OF ELECTRONICS & COMMUNICATION ENGINEERING
COLLEGE OF ENGINEERING TRIVANDRUM
2024-2026



CERTIFICATE

This is to certify that the Internship Report submitted by, **SHARANSUNDAR S** (TVE24ECRA13), to the APJ Abdul Kalam Technological University in partial fulfillment of the M.Tech. degree in Robotics and Automation is a bonafide record of the internship work carried out by him under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

Prof. SHAFEEK M
Assistant Professor
(Internship Co-ordinator)
Bosch Rexroth-CET
Centre of Excellence in
Automation Technologies
College of Engineering
Trivandrum

Dr. HARRIS P A
Professor and Head
Dept. of ECE
College of Engineering
Trivandrum

Abstract

The internship at the Bosch Rexroth–CET Center of Excellence in Automation Technologies, College of Engineering Trivandrum, provided comprehensive exposure to industrial automation, robotics, and process control systems. The program integrated theoretical briefings with practical demonstrations and hands-on training, enabling participants to apply academic knowledge in real-world automation scenarios.

Key activities included the implementation of a **pick-and-place project using the ABB IRB 120 manipulator**, integrating color sensing, limit switches, and conveyor automation for precise sorting operations. Gained experience in programming the IRC5 compact controller using RAPID, system-level coordination of robots and peripherals, and sensor-actuator interfacing.

The internship also involved **LEGO Mindstorms EV3 robotics** - assembled and programmed predefined models (Everstorm, Raptor, Tracker, Manipulator) and a custom conveyor system. These exercises reinforced understanding of fundamental robotics concepts, modular design, sensor integration, and programming logic.

Additionally, conducted **workshop sessions**, demonstrating both educational and industrial robotic systems, explaining robotics fundamentals such as types of robots, degrees of freedom, joints, workspaces, and forward and inverse kinematics. Exposure to **the Universal Process Control Trainer (UPCT-330)** provided practical understanding of industrial control loops, monitoring, PID tuning, and system calibration.

The internship fostered technical competency in robotics and automation, enhanced system integration skills, and developed critical problem-solving abilities. It also promoted collaboration, creativity, and adaptability, equipping with practical skills essential for future academic research or industrial applications in automation and intelligent systems.

Acknowledgement

I take this opportunity to express my deepest sense of gratitude and sincere thanks to everyone who helped us to complete this Internship successfully. I express my sincere thanks to **Dr. SURESH K** , Principal, and **Dr. HARRIS P A** , Head of Department, Electronics and Communication Engineering, College of Engineering Trivandrum for providing us with all the necessary facilities and support. I would like to express my sincere gratitude to **Prof. SHAFEEK M**, Internship Coordinator, Bosch Rexroth-CET Centre of Excellence in Automation Technologies, College of Engineering Trivandrum, for his support and co-operation.

Contents

Abstract	i
Acknowledgement	ii
List of Figures	vii
1 Overview of the Internship Program	1
2 Pick-and-Place Project Using ABB IRB 120	3
2.1 Introduction	3
2.1.1 Background and Motivation	3
2.1.2 Objectives of the Project	4
2.1.3 Scope of the Work	4
2.2 System Overview	5
2.2.1 Description of the ABB IRB 120 Manipulator	5
2.2.2 Overview of IRC5 Compact Controller	6
2.2.3 Overview of the Pick and Place Operation	7
2.2.4 Conveyor Mechanism using Aluminium Profile	7
2.2.5 Sensors Used in the System	8
2.2.6 Arduino-Based Control System	9
2.2.7 Role of RobotStudio Software	9
2.3 Operation of ABB IRB 120 Manipulator	10
2.3.1 Switching on the Main Power Box	10
2.3.2 Turning On the Controller	11
2.3.3 Activating the Pneumatic Gripper Air Compressor	11
2.3.4 Teach Pendant and IRC5 Controller Interface	12

2.3.5	Dead Man Switch Safety Mechanism	12
2.3.6	CMOS Battery Failure Error Display	13
2.3.7	Calibration Menu on Teach Pendant	13
2.3.8	Successful Calibration Display	14
2.3.9	Jogging Menu – Cartesian Movement	14
2.3.10	Jogging Menu – Joint Space Movement	15
2.3.11	Routines (Programs) Menu	15
2.3.12	Zero Position Code Execution	16
2.3.13	Teach Pendant I/O List Display	16
2.4	Hardware Components	17
2.4.1	ABB IRB 120 Robotic Arm	17
2.4.2	IRC5 Compact Controller	18
2.4.3	Sensor Modules	19
2.4.4	Conveyor System	20
2.4.5	Microcontroller Units	21
2.4.6	Optocoupler Circuit for Logic Isolation	22
2.5	Software and Programming	23
2.5.1	Software Stack Overview	23
2.5.2	RAPID Programming in the Teach Pendant	23
2.5.3	RobotStudio for Simulation and Debugging	24
2.5.4	Arduino Programming for Sensors and Conveyor Control	24
2.5.5	Synchronization Between Robot and Microcontrollers	25
2.5.6	Safety and Error Handling	26
2.6	System Integration	26
2.6.1	Overview	27
2.6.2	System Workflow	27
2.6.3	Signal Interfacing and Control	28
2.6.4	Power Supply Integration	28
2.6.5	Synchronization Logic	28
2.6.6	Mechanical Integration	29
2.6.7	Challenges and Solutions	31
2.7	Working Methodology	32

2.7.1	Introduction	32
2.7.2	Block Arrival and Detection	32
2.7.3	Color Detection and Classification	33
2.7.4	Interfacing with IRC5 Compact Controller	33
2.7.5	Robotic Pick-and-Place Operation	33
2.7.6	Tray Feedback and Verification	34
2.7.7	Conveyor Resumption and Cycle Reset	34
2.7.8	Timing and Synchronization	34
2.7.9	Error Handling and Robustness	35
2.8	Challenges and Solutions	35
2.8.1	Introduction	35
2.8.2	Conveyor Design Challenges	35
2.8.3	Gripping and End Effector Challenges	37
2.8.4	Workstation Stability	37
2.8.5	Housing for Electronics	38
2.8.6	Software and Logic Challenges	38
2.9	Testing and Results & Conclusion	39
2.9.1	Testing and Results	39
2.9.2	Conclusion	40
2.10	Future Work	41
2.10.1	Introduction	41
2.10.2	Advanced Object Recognition	41
2.10.3	Conveyor Enhancements	41
2.10.4	Gripper and End Effector Enhancements	42
2.10.5	System Intelligence and Industry 4.0 Readiness	42
2.10.6	Error Handling, Diagnostics, and Safety	42
2.10.7	Scalability and Flexible Production	43
3	LEGO Robotics Systems	44
3.1	Introduction to LEGO Mindstorms EV3	44
3.2	Assembly of Models	44
3.2.1	EV3 Models	44

3.2.2	Custom-Built Conveyor Mechanism	45
3.3	EV3 Controllers & Programming	46
3.3.1	Brick Programming Language in EV3 Classroom	46
3.3.2	Demonstrations of Control Logic	47
3.4	Hands-on Implementation	47
3.5	Challenges and Learnings	48
3.5.1	Challenges	48
3.5.2	Learnings	48
4	Workshop Sessions	49
4.1	Planning and Content of Sessions	49
4.2	Demonstrations Using LEGO Kits	49
4.3	Demonstrations of ABB IRB 120 Manipulator and Project Work . . .	50
4.4	Hands-on Exposure Given to Students	51
4.5	Feedback and Learning Outcomes	51
5	Universal Process Controller – Experimental Study	54
5.1	Introduction	54
5.2	Experimental Setup	54
5.3	Experiments Performed (Part IV – Control Loops)	55
5.4	Conclusion	59
6	Overall Internship Outcomes	60
7	Conclusion and Future Scope	63
	References	66

List of Figures

2.1	IRC5 Compact controller used to control IRB 120	6
2.2	Conveyor mechanism built with aluminium profile and stepper motor drive	8
2.3	IR sensor (left) and APDS-9960 color sensor (right) used for block detection	9
2.4	Main Power Box for ABB IRB 120 Manipulator.	10
2.5	IRC5 Compact Controller Powering Up.	11
2.6	Air Compressor for Pneumatic Gripper.	11
2.7	Teach Pendant and IRC5 Compact Controller.	12
2.8	Dead Man Switch on the Flex Pendant.	12
2.9	CMOS Battery Failure Error on Teach Pendant Display.	13
2.10	Calibration Menu on Teach Pendant.	13
2.11	Successful Calibration Confirmation on Teach Pendant.	14
2.12	Cartesian Movement in Jogging Menu.	14
2.13	Joint Space Movement in Jogging Menu.	15
2.14	Routines (Programs) Menu on Teach Pendant.	15
2.15	Zero Position Code Execution on ABB IRB 120.	16
2.16	Teach Pendant Display Showing Input/Output List.	16
2.17	ABB IRB 120 robotic arm equipped with pneumatic gripper	18
2.18	ABB IRC5 Compact controller used to control IRB 120	19
2.19	Conveyor built using aluminium profiles and stepper motor	21
2.20	ABB FlexPendant	24
2.21	3D modelled Mount for Buck Converter	30
2.22	Arduino and Color Sensor Mount	30

2.23	3D model of motor mount for stepper motor	30
2.24	Bearing Housing Mount for Conveyor Rollers	31
2.25	Initial wooden conveyor prototype	36
2.26	Redesigned conveyor using aluminium profile	37
3.1	LEGO MINDSTORMS EV3 kit and core components	45
3.2	EV3RSTORM (left) and R3PTAR (right).	46
3.3	TRACK3R (left) and Manipulator (right).	46
3.4	EV3 Classroom block program (e.g., line following).	47
4.1	Demonstration of LEGO EV3 Kits	50
4.2	Demonstration of ABB IRB 120 Manipulator and conveyor system . .	52
4.3	Student Interaction During Workshop	53
5.1	Univesal Process Control Trainer 330	56
5.2	Schematic of the Universal Process Control Trainer (UPCT-330). . . .	57

Chapter 1

Overview of the Internship Program

The internship was carried out at the **Bosch Rexroth-CET Center of Excellence in Automation Technologies**, located at the College of Engineering Trivandrum. The center is a collaborative initiative that provides students with a unique platform to acquire hands-on experience in advanced domains such as industrial automation, robotics, mechatronics, and process control. Its objective is to bridge the gap between academic learning and industrial practices by equipping students with practical skills aligned with the latest technological trends.

This internship was designed to combine theoretical foundations with practical applications, thereby strengthening both technical expertise and problem-solving capabilities. Over the course of the program, we were actively involved in multiple activities and projects that provided a holistic understanding of robotics and automation. The key components of the internship are summarized as follows:

- **Pick-and-Place Project:** Implementation of a robotic pick-and-place task using the ABB IRB 120 robotic arm integrated with a pneumatic gripper and external sensors. The project emphasized robot programming, sensor integration, and real-time control for executing industrial-relevant automation tasks.
- **LEGO Robotics:** Assembly and programming of different robotic models using LEGO Mindstorms EV3 kits. Predefined models such as *Everstorm*, *Raptor*, *Tracker*, and *Manipulator* were explored, along with the development of a custom-designed conveyor system. Programming was carried out in the **EV3 Classroom environment** using the LEGO Brick programming interface, thereby

providing an educational perspective to robotics learning.

- **Knowledge Transfer Sessions:** Engagement in hands-on training sessions for undergraduate workshop participants. These sessions included demonstrations and tutorials on robotics fundamentals, programming basics, and LEGO robot construction. This activity helped in enhancing both technical communication and teaching skills.
- **Study of Universal Process Controller:** A detailed study of the **Universal Process Controller** available in the lab, with emphasis on its role in process automation and control systems. The exposure to process controllers highlighted their significance in industrial operations.

Overall, the internship offered a multi-dimensional learning experience that not only involved technical execution but also collaboration, teamwork, and instructional roles. By working on industrial-grade systems alongside educational robotics kits, we gained valuable insights into both the professional and pedagogical aspects of robotics and automation. This combination of exposure enabled us to develop a balanced perspective of automation technologies applicable to both industry and academia.

Chapter 2

Pick-and-Place Project Using ABB IRB 120

2.1 Introduction

2.1.1 Background and Motivation

Automation is a cornerstone of modern manufacturing and industrial processes, enabling increased productivity, precision, and consistency while reducing human error and labor-intensive tasks. Among the various automation technologies, robotic manipulators play a vital role in executing repetitive operations such as material handling, assembly, and inspection. The ABB IRB 120 is a compact six-axis industrial robot widely used in academic research and industrial applications due to its precision, reliability, and ease of integration.

This project was undertaken as part of an industrial automation internship with the objective of developing a **color-based pick and place system** using the ABB IRB 120 robotic manipulator. The system is designed to detect colored square blocks on a conveyor, identify their color, and place them in corresponding trays. This type of application is especially relevant in automated sorting lines in industries such as packaging, pharmaceuticals, and electronics, where object categorization based on visual cues is required.

The project not only emphasizes the robotic manipulation aspect but also focuses on system integration involving sensor feedback, I/O interfacing, and conveyor control,

thereby providing a comprehensive understanding of how various components in an industrial cell work together.

2.1.2 Objectives of the Project

The main objectives of this project are:

- To study and understand the working of the **ABB IRB 120 robotic manipulator** and its **IRC5 Compact controller**.
- To design and implement a **pick and place system** that uses **color sensing** for sorting blocks into different trays.
- To build and automate a **conveyor system** using aluminium profiles and integrate it with the robot's operation.
- To explore and implement **I/O integration techniques** to allow communication between external sensors and the robot controller.
- To program the robot using **RAPID** language and configure it via the **FlexPendant**.
- To interface **Arduino microcontrollers** for controlling sensors and actuators, and to handle logic-level differences using **optocouplers**.
- To monitor and simulate the robot operation using **RobotStudio**.
- To propose improvements for future scalability using **OpenCV** and **ArUco markers** for advanced vision-based sorting.

2.1.3 Scope of the Work

This project encompasses the following:

- Mechanical setup and alignment of the ABB IRB 120 robot and the conveyor mechanism.
- Hardware integration of sensors (IR, color sensor, and limit switches) and actuators (conveyor motor, custom tray switches).

- Software programming using RAPID (for robot) and Arduino C/C++ (for microcontrollers).
- Interfacing and signal compatibility using optocouplers to connect 5V logic to 24V robot I/O modules.
- Implementation of a working pick and place cycle based on real-time sensor input.
- Simulation and debugging using ABB RobotStudio.
- Documentation of the integration process and suggestions for further enhancements using computer vision.

This project demonstrates the integration of industrial robotics with embedded systems and real-time automation control, providing valuable hands-on experience in multi-domain engineering required in modern smart factories.

2.2 System Overview

This section provides a detailed description of the key components and systems used in the development of the color-based pick and place automation project using the ABB IRB 120 robotic manipulator. The system is a hybrid integration of industrial robotics, embedded electronics, sensing systems, and mechanical automation elements. The overall functionality is realized through synchronized operation between the robot, the conveyor, and the external sensors.

2.2.1 Description of the ABB IRB 120 Manipulator

The ABB IRB 120 is a six-axis industrial robotic arm known for its compact size, high precision, and versatility. It has a payload capacity of 3 kg and a reach of 580 mm, making it suitable for small parts assembly, laboratory automation, and educational applications. The manipulator uses servo motors at each joint to allow movement along six degrees of freedom (DoF), enabling it to perform complex motions and reach various target positions.

The end-effector used in this project is a **pneumatic two-finger gripper**, which is ideal for grasping rigid square blocks. The gripper is controlled via digital I/O signals configured on the robot's controller. The robot is mounted securely on a base frame near the conveyor to allow for easy access to incoming blocks.

2.2.2 Overview of IRC5 Compact Controller

The IRB 120 is controlled by ABB's **IRC5 Compact Controller**, which provides all computational and control capabilities required for the robot. It supports ABB's proprietary **RAPID programming language** and comes with interfaces for digital and analog I/Os, safety relays, and communication ports.

Key features of the IRC5 Compact used in this project include:

- Digital I/O ports (24V logic) for sensor and actuator interfacing.
- Connectivity to ABB RobotStudio for offline programming, monitoring, and debugging.
- Real-time control via FlexPendant for jogging, editing programs, and setting variables.

Due to the controller's 24V I/O requirement, **optocouplers** were used to safely connect 5V logic signals from microcontrollers such as Arduino Uno and Mega.



Figure 2.1: IRC5 Compact controller used to control IRB 120
source:<https://new.abb.com/products/robotics/controllers/irc5-overview/irc5c>

2.2.3 Overview of the Pick and Place Operation

The primary task of the system is to:

1. Detect the presence of a block on a moving conveyor using IR sensors.
2. Identify the color of the block using the **APDS-9960 color sensor**.
3. Command the robot to pick up the block.
4. Sort and place the block into a predefined tray based on its color.

The robot executes the pick and place sequence through RAPID programming, responding to signals received from the conveyor system and sensors.

2.2.4 Conveyor Mechanism using Aluminium Profile

A custom conveyor mechanism was built using **aluminium extrusion profiles**, selected for their modularity, structural strength, and ease of prototyping. The conveyor is powered by a **stepper motor**, which allows for precise control of motion and positioning, important for stopping blocks at exact pickup points. The stepper motor is driven by an Arduino-based driver circuit.

Sensors mounted on the conveyor include:

- **IR sensors** to detect the presence of blocks at key positions.
- **APDS-9960 color sensor** to detect block color.
- **Limit switches** to define conveyor limits and ensure operational safety.

The conveyor is synchronized with the robot such that it pauses when a block reaches the pickup point and resumes only after sorting is completed.

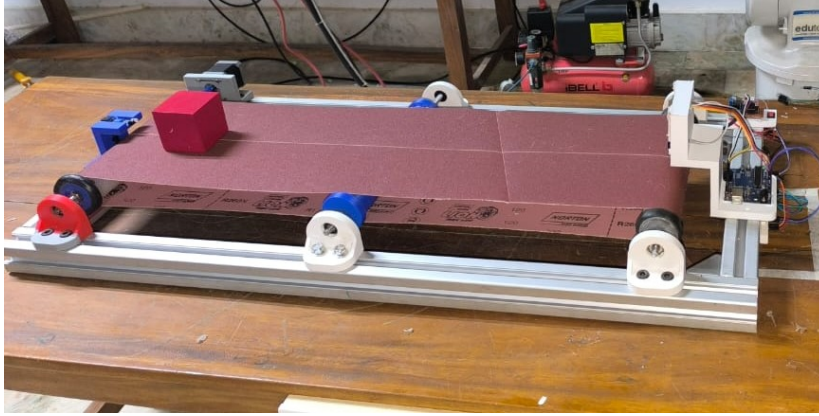


Figure 2.2: Conveyor mechanism built with aluminium profile and stepper motor drive

2.2.5 Sensors Used in the System

IR Sensor

The **infrared sensor** is used to detect the presence of objects (blocks) on the conveyor. It provides a digital HIGH/LOW signal based on reflection from the object's surface and is suitable for short-range, fast detection.

Color Sensor: APDS-9960

The **APDS-9960** sensor provides RGB color data and proximity sensing in a compact module. It communicates via the I²C protocol and is used to detect the color of each block. Based on calibrated RGB thresholds, the Arduino classifies blocks into different color categories and sends sorting commands to the robot controller.

Limit Switches (Custom)

Custom-built mechanical limit switches are placed at:

- The end of the conveyor to prevent overrun.
- Each sorting tray to provide feedback confirming the block has been placed.

These switches ensure safe operation and help the robot identify the status of each sorting cycle.

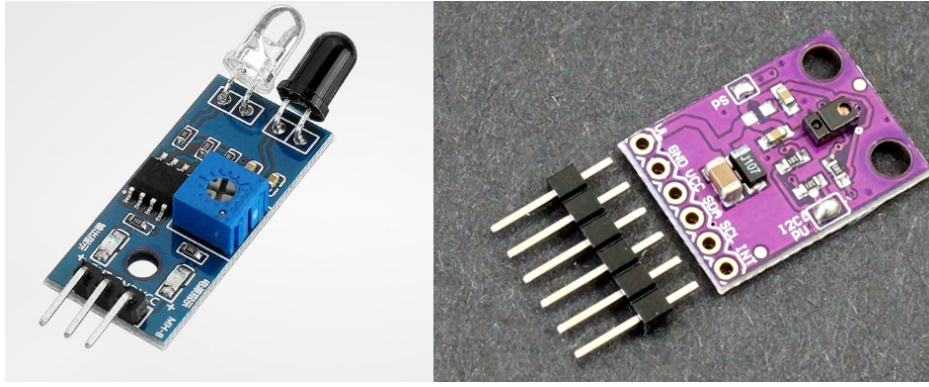


Figure 2.3: IR sensor (left) and APDS-9960 color sensor (right) used for block detection

2.2.6 Arduino-Based Control System

Two microcontrollers are used for control and integration:

- **Arduino Uno** handles sensor data collection (IR and APDS-9960).
- **Arduino Mega** manages conveyor control (stepper motor), tray limit switch inputs, and communicates with the robot via opto-isolated digital lines.

The Arduino boards operate at 5V logic and require level shifting to interface safely with the robot's 24V logic. For this, optocoupler-based circuits were designed and integrated into the control system.

2.2.7 Role of RobotStudio Software

ABB RobotStudio is used for simulation, programming, and real-time monitoring of the IRB 120 manipulator. Key functionalities include:

- Offline simulation of robot movement and path planning.
- Visualization of I/O signals and live system variables.
- Debugging of RAPID programs and cycle testing without physical hardware.

RobotStudio ensured the pick and place logic was optimized before deployment, minimizing trial-and-error on the live system.

This section presented a complete overview of all mechanical, electrical, and software components used in the project. The following sections will explore the individual modules in detail, including system integration, real-time logic, and programming.

2.3 Operation of ABB IRB 120 Manipulator

This section details the step-by-step procedure for powering up, calibrating, and operating the ABB IRB 120 manipulator. Each subsection includes an explanation of the operation and a placeholder for the relevant image.

2.3.1 Switching on the Main Power Box

The operation begins by turning on the main power box of the manipulator. This supplies electrical power to the robot controller and associated systems.



Figure 2.4: Main Power Box for ABB IRB 120 Manipulator.

2.3.2 Turning On the Controller

After the main power is on, the IRC5 compact controller should be powered up. This initializes the robot's control system and prepares it for communication with the teach pendant.



Figure 2.5: IRC5 Compact Controller Powering Up.

2.3.3 Activating the Pneumatic Gripper Air Compressor

The pneumatic gripper requires a pressurized air supply. Ensure that the air compressor is turned on to provide the necessary air for gripping operations.



Figure 2.6: Air Compressor for Pneumatic Gripper.

2.3.4 Teach Pendant and IRC5 Controller Interface

The teach (flex) pendant provides the human-machine interface for programming and operating the manipulator. The IRC5 controller communicates with the pendant to display system status and execute commands.



Figure 2.7: Teach Pendant and IRC5 Compact Controller.

2.3.5 Dead Man Switch Safety Mechanism

The dead man switch on the flex pendant ensures operator safety. The robot only moves when the switch is pressed, preventing unintended motions.



Figure 2.8: Dead Man Switch on the Flex Pendant.

2.3.6 CMOS Battery Failure Error Display

Sometimes, the pendant displays a CMOS battery failure warning. This error indicates that the internal clock battery needs attention before proceeding with operations.

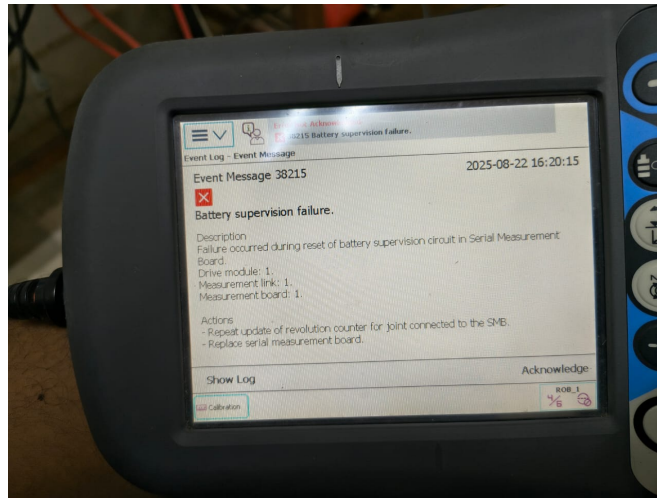


Figure 2.9: CMOS Battery Failure Error on Teach Pendant Display.

2.3.7 Calibration Menu on Teach Pendant

Calibration is required each time the manipulator is powered on due to CMOS battery failure. The revolution counters of each joint must be manually updated to ensure accurate motion control.

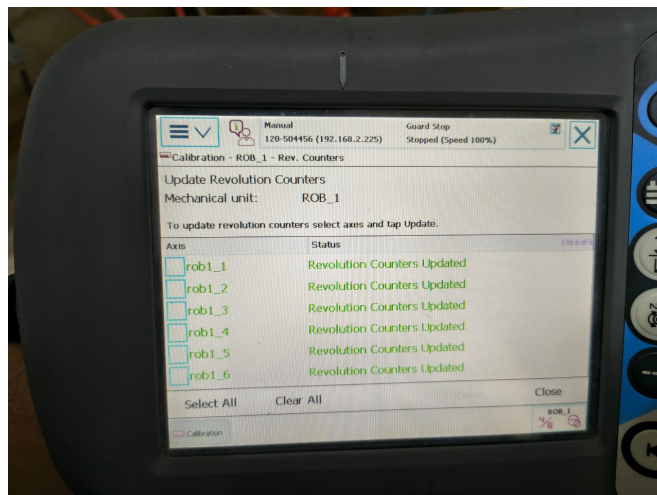


Figure 2.10: Calibration Menu on Teach Pendant.

2.3.8 Successful Calibration Display

After updating the revolution counters, the pendant confirms successful calibration. This ensures that the robot's joint positions are correctly referenced.

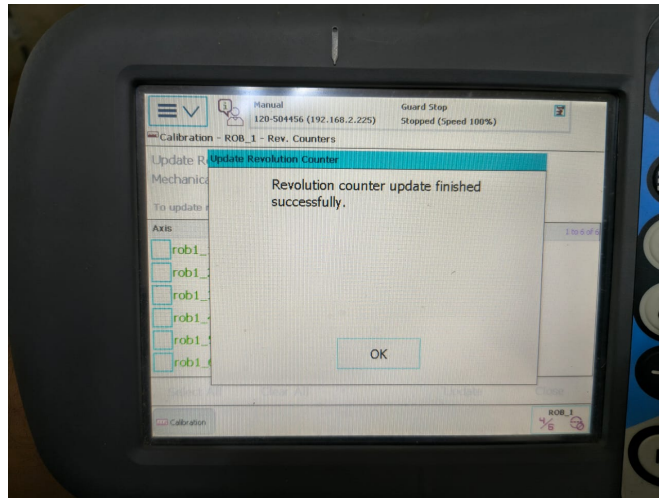


Figure 2.11: Successful Calibration Confirmation on Teach Pendant.

2.3.9 Jogging Menu – Cartesian Movement

The jogging menu allows movement of the robot in Cartesian space (X, Y, Z axes) for positioning and testing.

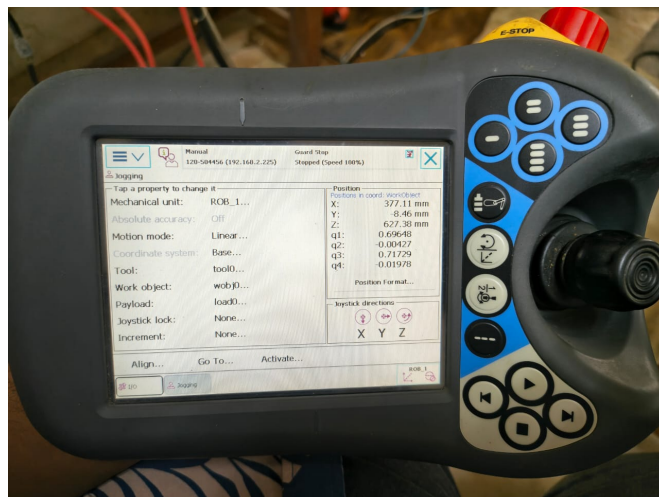


Figure 2.12: Cartesian Movement in Jogging Menu.

2.3.10 Jogging Menu – Joint Space Movement

Joint space jogging moves the robot by individually controlling each joint. This is useful for fine positioning and calibration.



Figure 2.13: Joint Space Movement in Jogging Menu.

2.3.11 Routines (Programs) Menu

The routines menu on the teach pendant displays the available programs. It allows starting, editing, or creating new routines for automated tasks.

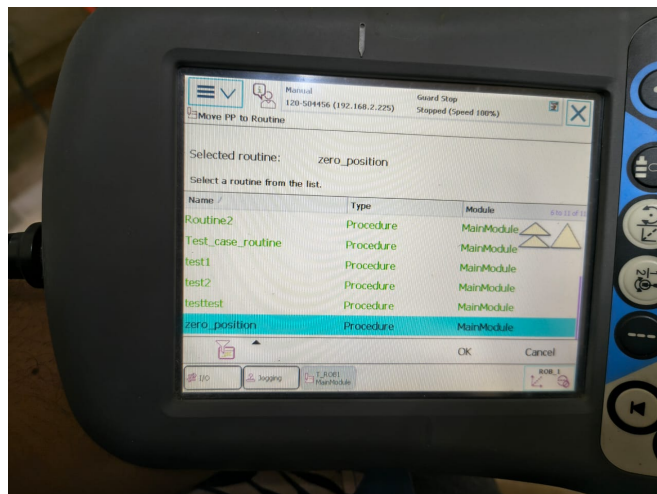


Figure 2.14: Routines (Programs) Menu on Teach Pendant.

2.3.12 Zero Position Code Execution

The zero position code resets the manipulator to a predefined home position. This ensures consistent starting coordinates for automated operations.

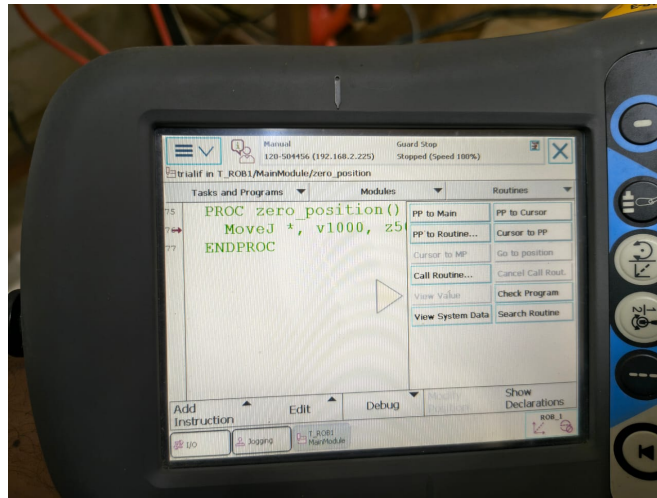


Figure 2.15: Zero Position Code Execution on ABB IRB 120.

2.3.13 Teach Pendant I/O List Display

The teach pendant provides an interface to monitor and configure the input and output signals of the robot system. The I/O list display shows all digital and analog inputs and outputs connected to the IRC5 controller, allowing verification of sensor and actuator states and facilitating troubleshooting during operations.

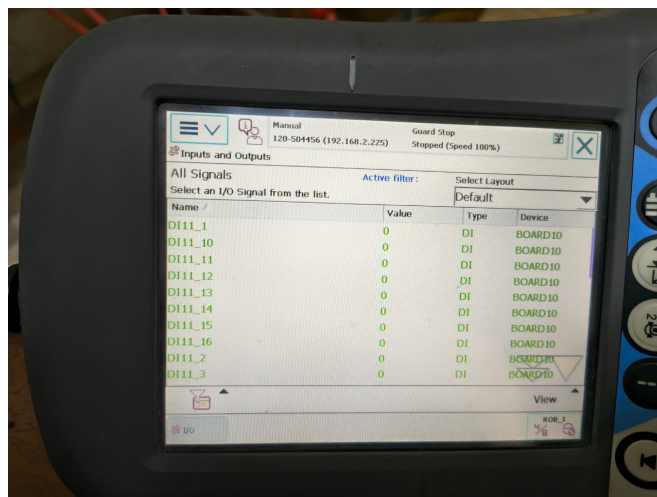


Figure 2.16: Teach Pendant Display Showing Input/Output List.

2.4 Hardware Components

This section discusses the various hardware components employed in the color-based pick and place automation system. The project integrates industrial-grade robotics with embedded systems and custom mechanical design. Each component is carefully selected for its functionality, compatibility, and reliability in real-time industrial tasks.

2.4.1 ABB IRB 120 Robotic Arm

The ABB IRB 120 is a six-axis industrial robot known for its lightweight structure, high precision, and versatility in laboratory and light manufacturing environments. It has a payload capacity of 3 kg and a maximum reach of 580 mm, making it ideal for applications such as pick and place, small parts assembly, and material handling.

Key specifications:

- Degrees of Freedom: 6
- Payload: 3 kg
- Maximum Reach: 580 mm
- Repeatability: ± 0.01 mm
- Weight: 25 kg

In this project, the robot is responsible for performing pick and place operations. The robot arm is fitted with a **pneumatic two-finger gripper** capable of gripping small square blocks made of plastic or wood. The robot executes its tasks by following programmed waypoints using RAPID language via the teach pendant.



Figure 2.17: ABB IRB 120 robotic arm equipped with pneumatic gripper

2.4.2 IRC5 Compact Controller

The **IRC5 Compact** is ABB's controller used to manage the IRB 120 robot. It is a scaled-down version of the standard IRC5 but retains all key functionalities such as RAPID programming, motion control, and digital I/O management.

It provides:

- Real-time robot control using RAPID language
- I/O ports for digital signal communication (24V logic)
- Connectivity with RobotStudio software for simulation and online monitoring
- FlexPendant interface for manual teaching and debugging

The controller interfaces with external hardware via digital I/Os. Since it operates on 24V logic, special care is taken using optocoupler circuits when integrating 5V logic-based devices like Arduino.



Figure 2.18: ABB IRC5 Compact controller used to control IRB 120

2.4.3 Sensor Modules

IR Sensor

The **infrared sensor module** is used to detect the presence of blocks on the conveyor. It operates based on infrared reflection. When a block interrupts the IR beam, the sensor's output changes state.

Features:

- Operates at 3.3V to 5V
- Digital output (HIGH/LOW)
- Adjustable sensitivity using onboard potentiometer
- Fast response and low cost

These sensors are mounted at key locations:

- At the pickup zone (to detect block arrival)

- At the entry and exit points of the conveyor

APDS-9960 Color Sensor

The **APDS-9960** is a multi-functional sensor capable of detecting ambient light, RGB values, proximity, and gestures. In this project, it is used for color detection of blocks. It communicates with the Arduino over the I²C protocol.

Working Principle:

- Contains four photodiodes filtered for red, green, blue, and clear light
- Captures intensity values for each channel
- Arduino maps the RGB values to predefined color categories

This sensor allows real-time block classification, which is critical to determine the sorting tray for the pick and place operation.

Custom Limit Switches

Custom-built mechanical limit switches are employed for:

- Conveyor end detection (to prevent overrun)
- Tray feedback (to confirm placement of blocks)

These switches provide HIGH/LOW digital signals to the Arduino Mega, which forwards the information to the robot controller.

2.4.4 Conveyor System

The conveyor mechanism is a key subsystem built using aluminium extrusion profiles (30×30). It consists of:

- A flat conveyor belt
- A **stepper motor** for actuation
- Bearings and guide rails for smooth movement

The use of a stepper motor provides precise control over belt movement. The motor is controlled via Arduino using a motor driver (e.g., A4988 or ULN2003), enabling:

- Start-stop control at pickup points
- Variable speed control
- Direction control

The conveyor works in sync with the robot: when a block is detected, the conveyor halts until the robot completes the pick operation.

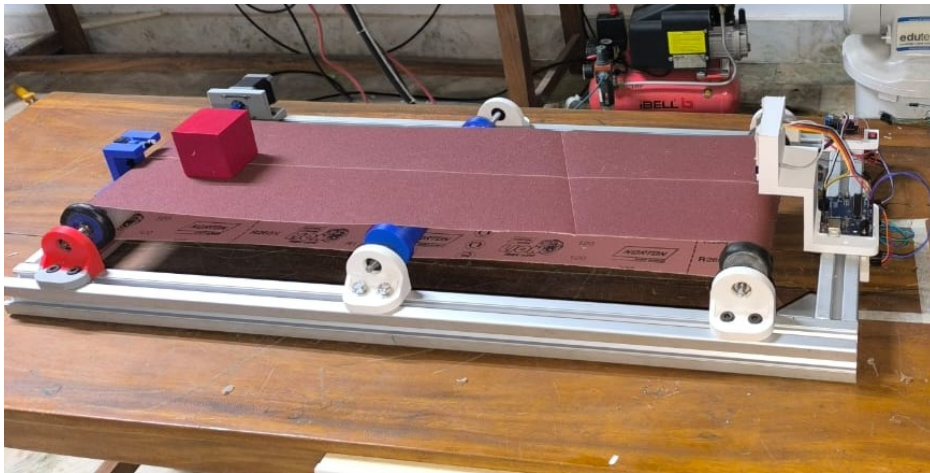


Figure 2.19: Conveyor built using aluminium profiles and stepper motor

2.4.5 Microcontroller Units

Arduino Uno

The Arduino Uno is an ATmega328P-based microcontroller used in this project to:

- Acquire data from the APDS-9960 color sensor
- Monitor the IR sensor
- Send color classification result to Arduino Mega

It is powered via USB or 5V external supply and communicates via serial/I2C interfaces.

Arduino Mega

The Arduino Mega 2560, featuring more I/O pins and memory than Uno, is used for:

- Controlling the stepper motor (through motor driver)
- Reading tray limit switch feedback
- Sending status signals to the IRC5 controller via optocouplers

By separating tasks between Uno and Mega, the system achieves modularity and efficient parallel processing.

2.4.6 Optocoupler Circuit for Logic Isolation

Interfacing 5V Arduino outputs with the 24V digital inputs of the IRC5 Compact controller poses a risk of damaging the microcontroller. To resolve this, an **optocoupler circuit** is used.

Working:

- The Arduino output turns on an internal LED inside the optocoupler
- The light activates a phototransistor on the output side
- The isolated transistor completes the circuit on the 24V side safely

Components used:

- PC817 optocoupler
- 330 Ω resistor for LED current limiting
- Pull-up resistor for output stage

This ensures that any signal sent from the Arduino to the robot's I/O module is electrically isolated and safe.

In summary, this section outlined each hardware component's purpose and integration in the system. The combination of industrial robotics, microcontroller logic, sensor feedback, and motion control forms a cohesive and scalable automation solution.

2.5 Software and Programming

This section outlines the software architecture and programming logic implemented in the project. It elaborates on the use of RAPID programming for robot control, Arduino programming for sensor and actuator interfacing, and RobotStudio for simulation and debugging. The synchronization mechanisms between the robot, sensors, and conveyor are also discussed in detail.

2.5.1 Software Stack Overview

The system software is composed of multiple platforms that operate collaboratively:

- **RAPID Language:** Used for robot control programming via the teach pendant.
- **Arduino IDE (C/C++):** Used for controlling sensors, stepper motor, and signal processing.
- **RobotStudio:** Used for simulation, debugging, and offline programming.
- **Digital I/O Communication:** Used for inter-device communication between robot and Arduino via optocouplers.

2.5.2 RAPID Programming in the Teach Pendant

RAPID is ABB's proprietary language used for programming robot motion and logic through the teach pendant. It is event-driven and supports real-time I/O monitoring, motion instructions, and variable declarations.

The robot's workflow was implemented using 'PROC' routines and motion commands like 'MoveJ', 'MoveL', and I/O instructions like 'WaitDI', 'SetDO'.

Listing 2.1: Sample RAPID Code for Pick and Place

```
WaitDI StartSensor, 1;  
MoveJ Home, v100, z10, tool0;  
MoveL PickPos, v50, fine, tool0;  
SetDO GripperClose, 1;  
WaitTime 0.5;
```

```
MoveL PlaceRed, v50, fine, tool0;  
SetDO GripperClose, 0;
```



Figure 2.20: ABB FlexPendant

Source: <https://quizlet.com/423610420/abb-robot-jogging-modes-flash-cards/>

2.5.3 RobotStudio for Simulation and Debugging

RobotStudio provides a 3D virtual environment to simulate and debug robot programs.

It enables verification of:

- Tool paths and workobject definitions
- Collision detection and workspace constraints
- Digital signal mapping and monitoring
- Execution of RAPID routines before uploading to hardware

It was instrumental in optimizing robot motion sequences without repeated physical trials.

2.5.4 Arduino Programming for Sensors and Conveyor Control

Two Arduino boards were programmed using Arduino IDE:

- **Arduino Uno:** Acquired data from APDS-9960 color sensor and IR sensor.
- **Arduino Mega:** Controlled stepper motor, read limit switches, and managed signal exchange with IRC5 controller via optocouplers.

Sensor data was processed and used to generate digital HIGH/LOW outputs that triggered robot actions. Stepper motors were driven using pulse signals generated through digital outputs.

Listing 2.2: Arduino Code for Color Detection using APDS-9960

```
#include <Wire.h>
#include <Adafruit_APDS9960.h>
Adafruit_APDS9960 apds;

void setup() {
  Serial.begin(9600);
  apds.begin();
}

void loop() {
  uint16_t r, g, b, c;
  apds.getColorData(&r, &g, &b, &c);
  if (r > g && r > b) {
    Serial.println("RED");
    digitalWrite(RED_PIN, HIGH);
  }
  delay(100);
}
```

2.5.5 Synchronization Between Robot and Microcontrollers

The system relied on real-time synchronization between the robot and the Arduino Mega using digital input/output lines connected through optocouplers.

- Arduino sends a HIGH signal to notify the robot about block readiness.

- Robot completes the pick and place routine and toggles a done signal.
- Arduino reads tray switch feedback and resumes conveyor movement.

This ensured seamless coordination and prevented logic mismatches or timing conflicts.

2.5.6 Safety and Error Handling

Basic error handling logic was incorporated to enhance reliability and protect hardware:

- Conveyor stops if block is not detected within timeout.
- Robot halts if tray limit switch fails to trigger after placement.
- Status LEDs indicate normal operation, tray full, or fault states.
- Debounce logic prevents false positives from mechanical switches.

Planned future upgrades include:

- Use of ArUco markers and OpenCV for accurate object localization
- Integration of serial communication from Arduino to robot via fieldbus

Summary

The project utilized multiple programming environments to control sensors, actuators, and robot behavior. Each software component was modularly developed and tested for robust real-time automation. The system's flexibility allows for future expansion and integration with vision-based techniques.

2.6 System Integration

This section describes how the various hardware and software modules were combined to form a unified, functioning system. Integration focused on the coordinated operation of sensors, actuators, controllers, and the robot, with special attention to logic synchronization, power interfacing, and physical arrangement.

2.6.1 Overview

The integration brings together several subsystems:

- ABB IRB 120 robotic manipulator
- IRC5 Compact controller
- Arduino Uno and Mega microcontrollers
- Conveyor system actuated by a stepper motor
- Sensor suite: IR sensor, APDS-9960 color sensor, and tray limit switches

These subsystems communicate and coordinate actions through digital I/O signals, serial communication, and synchronized programming routines.

2.6.2 System Workflow

The system follows a clearly defined sequence of operations to execute the pick and place routine:

1. Conveyor transports a colored block to the sensing zone.
2. IR sensor detects the block's presence.
3. APDS-9960 color sensor classifies the block as red, green, or blue.
4. Arduino Uno transmits the color classification to Arduino Mega.
5. Arduino Mega pauses the conveyor and sends a digital HIGH signal to the robot via optocoupler.
6. The robot picks the block and places it into the designated tray.
7. Tray limit switch sends confirmation to Arduino Mega.
8. Conveyor resumes operation to bring in the next block.

2.6.3 Signal Interfacing and Control

The IRC5 Compact controller operates at 24V logic, while Arduino operates at 5V. To prevent voltage incompatibility, optocoupler circuits are used. These provide:

- Electrical isolation between the robot and Arduino
- Safe logic level shifting
- Protection from back voltage spikes

Each trigger signal from Arduino Mega is passed through a PC817 optocoupler before reaching the robot's digital input terminal.

2.6.4 Power Supply Integration

The system operates on three voltage levels:

- 5V — for Arduino boards and sensors
- 12V — for stepper motor and driver
- 24V — for IRC5 Compact and optocoupler output circuits

To ensure reliable digital communication, all grounds were connected to a common reference. Proper current ratings were maintained for each component, and voltage regulators were used where necessary.

2.6.5 Synchronization Logic

Precise synchronization is required between robot and peripheral components. The logic implemented ensures:

- The robot executes motion only when a block is detected and classified.
- Conveyor movement halts during robot operation.
- Robot resumes only after receiving tray switch confirmation.
- Logic signals are not missed or misread due to bouncing or timing issues.

Timing conditions are handled in:

- RAPID code (using `WaitDI` and `SetDO`)
- Arduino Mega code (using `digitalRead/digitalWrite` with debounce logic)

2.6.6 Mechanical Integration

The physical layout and mounting of the system were designed with careful consideration, as they play a crucial role in ensuring the overall functionality and reliability of the robotic setup. Proper integration was necessary to achieve the following objectives:

- Accurate positioning of blocks beneath the robot's gripper.
- Unobstructed and smooth movement of both the conveyor system and the robot arm.
- Reliable placement of blocks into the designated trays without misalignment.

To achieve this, sensor holders, mechanical stoppers, and tray positions were aligned precisely using the robot's workobject coordinate system. This alignment ensured consistency in pick-and-place operations and minimized positional errors during repeated trials.

In addition to the above, several custom mounts were designed and 3D printed to integrate the sensors and electronic components with the conveyor system. These include:

- A dedicated mount for the buck converter, enabling stable and safe placement on the conveyor structure.
- An Arduino and color sensor mount, which ensured both mechanical stability and correct orientation of the color sensor for accurate block detection.
- An IR sensor mount, designed to hold the sensor at the appropriate distance from the conveyor surface to reliably detect the presence of blocks.
- A bearing housing mount, used to secure the bearings that support the rollers, thereby improving the conveyor's operational smoothness.

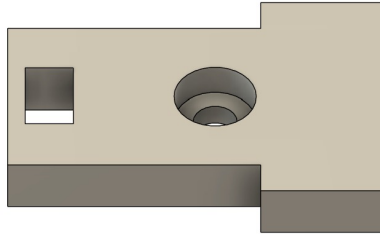


Figure 2.21: 3D modelled Mount for Buck Converter

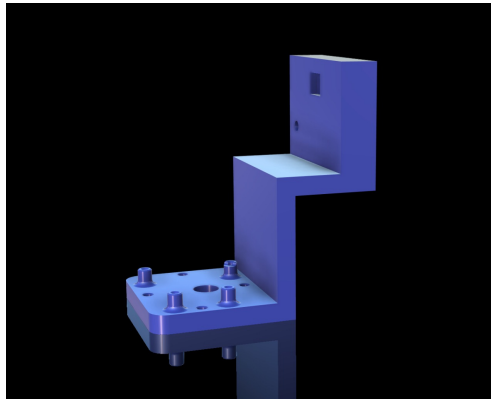


Figure 2.22: Arduino and Color Sensor Mount

- Motor mount for holding the stepper motor of the conveyor.

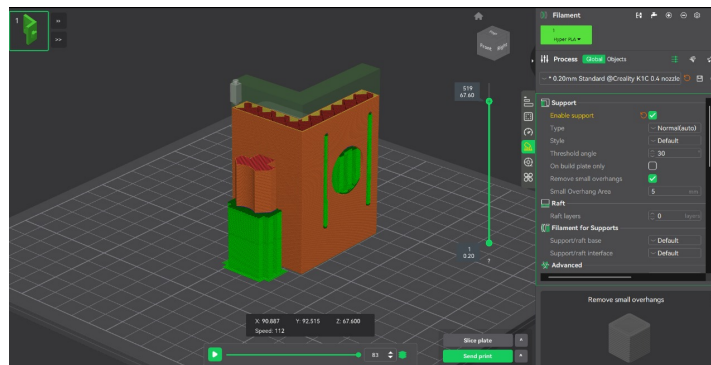


Figure 2.23: 3D model of motor mount for stepper motor

In addition to these mounts, an idle roller and a T-nut for tightening into the aluminum extrusion frame were also designed and 3D printed. These components, although not illustrated here, contributed significantly to the mechanical robustness and modularity of the conveyor assembly.

Overall, the custom-designed mounts and additional mechanical components ensured seamless integration of electrical and mechanical subsystems, while maintaining

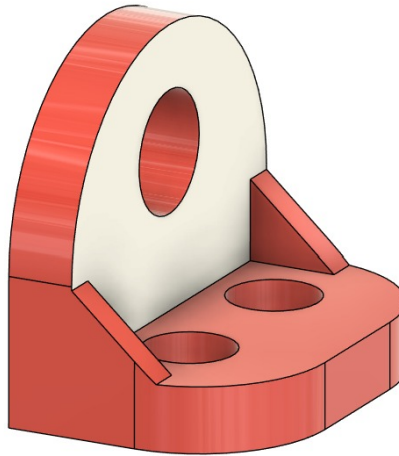


Figure 2.24: Bearing Housing Mount for Conveyor Rollers

the compactness and efficiency of the prototype.

2.6.7 Challenges and Solutions

Several challenges were faced during the integration phase:

- **Voltage mismatch:** Resolved using custom optocoupler circuits.
- **Color misclassification:** Calibrated APDS-9960 in controlled lighting.
- **Positioning errors:** Mechanical stoppers and software position offsets were added.
- **Switch bouncing:** Software debounce routines implemented in Arduino code.

Summary

The system integration phase successfully brought together the robotic manipulator, control electronics, sensors, and motion mechanisms. Careful design of signal flow, power distribution, and mechanical alignment ensured reliable and repeatable performance of the automated sorting system.

2.7 Working Methodology

This section outlines the complete workflow and logic behind the color-based pick and place operation using the ABB IRB 120 manipulator. It includes sensor processing, robotic coordination, timing synchronization, and error handling.

2.7.1 Introduction

The automation sequence is driven by real-time coordination between sensors, micro-controllers, and the robot. A finite state machine governs transitions through detection, classification, robotic action, and reset. The system ensures deterministic behavior, accuracy, and reliability.

2.7.2 Block Arrival and Detection

The IR sensor detects the presence of a block on the conveyor:

- Stepper-driven conveyor stops immediately.
- Arduino Uno is signaled to begin color analysis.

2.7.3 Color Detection and Classification

The APDS-9960 sensor reads RGB values. Arduino Uno processes these and categorizes the block into:

- Red
- Green
- Blue
- Yellow

The result is sent to Arduino Mega, which handles system control logic.

2.7.4 Interfacing with IRC5 Compact Controller

Color identification triggers digital outputs to the IRC5 controller via optocouplers. Each color has a dedicated 24V-compatible input on the robot controller. RAPID programs monitor inputs using `WaitDI` to determine action.

2.7.5 Robotic Pick-and-Place Operation

The ABB robot:

1. Moves to pick position
2. Closes the gripper
3. Moves to the respective tray
4. Opens gripper and releases block
5. Returns to home

Motion commands are implemented in RAPID using `MoveJ`, `MoveL`, `SetDO`, and `WaitTime`.

2.7.6 Tray Feedback and Verification

Each tray contains a limit switch:

- Confirms successful block placement
- Sends signal back to Arduino Mega
- System proceeds only upon confirmation

2.7.7 Conveyor Resumption and Cycle Reset

Once placement is verified:

- All signals to the robot are reset
- Conveyor resumes
- System returns to idle state

2.7.8 Timing and Synchronization

The workflow follows a finite state machine:

- IDLE
- BLOCK DETECTED
- COLOR CLASSIFIED
- ROBOT TRIGGERED
- BLOCK PLACED
- FEEDBACK RECEIVED
- RESET

2.7.9 Error Handling and Robustness

To ensure robustness:

- Tray timeout errors halt the system safely
- Invalid color detection is flagged and skipped
- IR sensor debounce prevents ghost triggers
- Manual override switch allows emergency stop

Summary

This methodology enables reliable color-based block sorting by integrating sensing, logic, motion control, and feedback. The modular design supports scalability, future upgrades, and adaptation to other object properties.

2.8 Challenges and Solutions

This section highlights the difficulties encountered while developing the color-based pick-and-place automation system and the solutions implemented to address them.

2.8.1 Introduction

Building the pick-and-place system involved overcoming challenges across the mechanical, electrical, and software domains. Iterative design and testing enabled the team to address these challenges systematically.

2.8.2 Conveyor Design Challenges

The initial conveyor was constructed using 4 mm sheets and wood. While simple, this design suffered from:

- Poor mechanical rigidity
- Frequent misalignment

- Difficulty integrating sensors
- Susceptibility to damage from robot interactions

Damage occurred when gripping failures caused blocks to shift unexpectedly, damaging the wooden conveyor.

Solution: An improved conveyor was built using aluminium profiles, providing:

- Higher mechanical stability
- Precise alignment
- Modular mounts for sensors and electronics



Figure 2.25: Initial wooden conveyor prototype

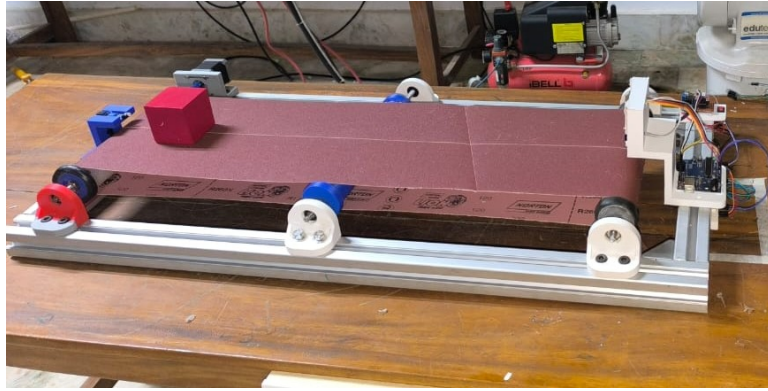


Figure 2.26: Redesigned conveyor using aluminium profile

2.8.3 Gripping and End Effector Challenges

The ABB IRB 120's gripper had limited gripping area, and the wooden blocks showed:

- Size variation
- Slight surface warping
- Low friction for gripping

This resulted in frequent pickup failures and collisions with the conveyor.

Solution:

- Used calibrated block sizes with better surface finishing
- Introduced a 3D-printed orientor to align blocks before picking

2.8.4 Workstation Stability

The workbench used initially lacked anchoring, causing:

- Vibrations during conveyor movement
- Misalignments affecting block pickup

Solution:

- Anchored the bench to a solid foundation
- Tuned conveyor speed profiles

2.8.5 Housing for Electronics

Arduino boards and sensors were initially mounted without protection, leading to:

- Cable clutter
- Risk of damage
- Difficult troubleshooting

Solution: Custom 3D-printed enclosures were developed for:

- Housing Arduino Mega and Uno
- Protecting sensors
- Neatly routing cables

2.8.6 Software and Logic Challenges

Software integration issues included:

- Timing mismatches between signals
- Color misclassification due to lighting changes
- IR sensor glitches

Solution:

- Implemented finite state machines
- Added debounce logic in Arduino code
- Calibrated sensors under uniform lighting
- Tuned RAPID code WaitDI timing conditions

Summary

Through iterative design and debugging, the team transformed the system from an unstable prototype to a robust, fully functional platform. Mechanical, electrical, and software challenges were addressed systematically to achieve a reliable automated pick-and-place system.

2.9 Testing and Results & Conclusion

2.9.1 Testing and Results

The developed color-based pick-and-place system was subjected to systematic testing to evaluate its accuracy, reliability, synchronization, and cycle time performance.

Testing Methodology

Testing focused on:

- Color detection accuracy (APDS-9960)
- Gripping and placement consistency (ABB IRB 120)
- Synchronization between conveyor (Arduino) and robot controller (IRC5 Compact)
- Cycle time analysis for each operation

Blocks of varying colors were placed on the conveyor. The system detected color, executed the pick-and-place operation, and sorted the blocks into designated trays. Multiple cycles were conducted to verify consistency.

Observations and Results

Table 2.1: Performance Summary of System Testing

Test Parameter	Expected	Observed	Remarks
Color Detection	Accurate under fixed lighting	Accurate for all colors	Occasional noise with glossy surfaces; mitigated through calibration.
Gripping Consistency	> 95% success rate	97% success rate	Failures mainly due to block misalignment.
Placement Accuracy	Correct tray placement	100% accurate placement	Verified through repeated tests.
Synchronization	Smooth conveyor-robot handoff	Achieved without delays	Efficient I/O signaling established.

System Performance Summary

- Stable, repeatable performance achieved.
- Accurate sorting post-calibration.
- Effective mechanical and software synchronization.
- Fine-tuning of sensors and conveyor improved reliability.

2.9.2 Conclusion

The internship project successfully delivered a reliable color-based pick-and-place automation system integrating ABB IRB 120, microcontroller-based sensors, and a conveyor mechanism.

Key outcomes include:

- Mastery of ABB RobotStudio and RAPID programming.
- Successful hardware-software interfacing between industrial and embedded systems.
- Resolution of mechanical, electrical, and software challenges through systematic refinement.
- Proven sorting reliability validated through structured testing.

The project demonstrates how industrial robotics can be effectively paired with cost-efficient sensor systems to achieve automation objectives, providing a solid foundation for future enhancements towards more intelligent and flexible automation solutions.

2.10 Future Work

This section discusses opportunities to expand and enhance the developed color-based pick-and-place system toward a more advanced and production-ready robotic cell.

2.10.1 Introduction

The prototype demonstrates solid color-based sorting capability. However, its potential can be expanded significantly with improvements in sensing, actuation, control intelligence, and safety features to meet industrial standards.

2.10.2 Advanced Object Recognition

Future development may involve:

- Vision-based detection using OpenCV with industrial cameras
- ArUco marker tracking for position and orientation accuracy
- CNN-based models to classify complex patterns beyond simple colors

Such improvements could greatly enhance classification robustness under variable lighting conditions.

2.10.3 Conveyor Enhancements

Proposed conveyor upgrades include:

- Servo-based conveyor drives with precise speed control
- Industrial belt standards with emergency stop features
- Modular extensions to handle higher throughput
- Smart conveyor zoning for simultaneous multiple block handling

These improvements would make the conveyor suitable for industrial production.

2.10.4 Gripper and End Effector Enhancements

Future systems could include:

- Adaptive soft grippers
- Integrated force-sensing
- Modular, 3D-printed gripper fingertips
- Electrically actuated variable-grip grippers

This would increase the range of objects that can be handled safely.

2.10.5 System Intelligence and Industry 4.0 Readiness

To modernize the system further:

- Integrate industrial PLCs for higher-level control
- Add IoT dashboards for remote supervision and data collection
- Incorporate predictive maintenance tools
- Maintain a traceable parts database

2.10.6 Error Handling, Diagnostics, and Safety

Further error handling may include:

- Database-based error logging with timestamps
- Watchdog timers to handle microcontroller failures
- Audio-visual alarms for operators
- Safety curtains or interlocks for human-robot coexistence
- Self-test procedures on startup

2.10.7 Scalability and Flexible Production

A scalable system might:

- Sort based on size, shape, or identity using RFID
- Integrate multiple robots in parallel
- Support fast reprogramming of tray positions and sorting rules
- Allow quick setup changes for different product lines

This would transform the prototype into a flexible manufacturing cell.

Summary

By pursuing these future work directions, the prototype can evolve into an advanced, intelligent, flexible, and safe industrial automation solution, well-aligned with Industry 4.0 practices and modern smart factory demands.

Chapter 3

LEGO Robotics Systems

3.1 Introduction to LEGO Mindstorms EV3

LEGO MINDSTORMS® EV3 combines LEGO construction elements with motors, sensors, and an intelligent controller (P-Brick) to build and program autonomous robots. The platform supports app-based programming and remote control, and provides official build guidance for multiple robots, making it suitable for rapid prototyping and classroom demonstrations.

3.2 Assembly of Models

To explore different mechanisms and behaviors, both standard EV3 builds and a custom mechanism were assembled.

3.2.1 EV3 Models

The following models were built and evaluated for motion, sensing, and task execution:

- **EV3RSTORM** — A humanoid robot demonstrating coordinated motion and interactive behaviors.
- **R3PTAR** — A serpentine -inspired robot emphasizing balance and dynamic movement.



Figure 3.1: LEGO MINDSTORMS EV3 kit and core components

Source: <https://www.lego.com/en-in/themes/mindstorms/ev3>

- **TRACK3R** — A wheeled mobile robot that tracks and follows predefined tracks.
- **Manipulator (custom adaptation)** — A simplified arm for pick-and-place tasks using EV3 motors and sensors.

3.2.2 Custom-Built Conveyor Mechanism

A compact conveyor mechanism was designed using EV3 structural elements and driven by EV3 motors. The conveyor supported object transport and integrated basic sensing to trigger motion and stop conditions, emulating a scaled industrial material-handling line.

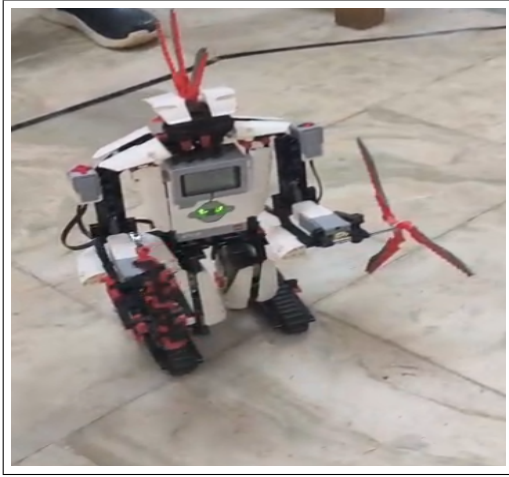


Figure 3.2: EV3RSTORM (left) and R3PTAR (right).

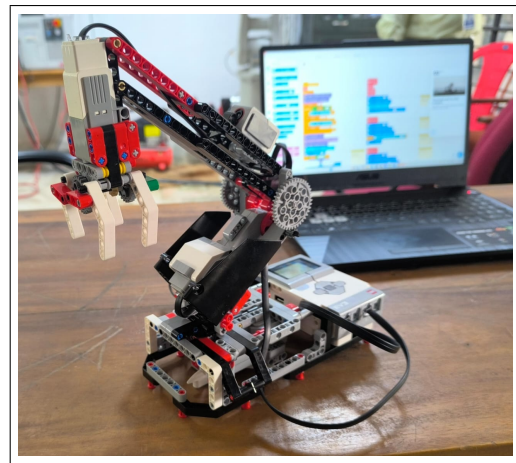
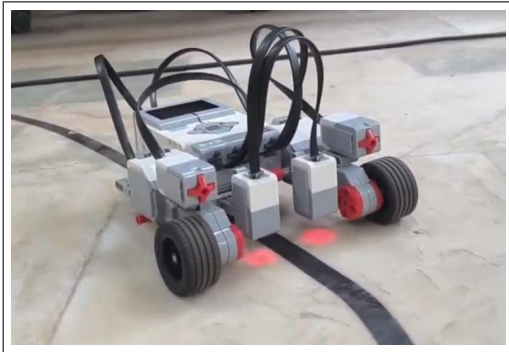


Figure 3.3: TRACK3R (left) and Manipulator (right).

3.3 EV3 Controllers & Programming

The EV3 system is controlled by the programmable **EV3 P-Brick**, with motor/sensor ports and onboard execution of user programs. For this internship, programs were created using the *EV3 Classroom* (block-based) environment; official LEGO apps such as the *EV3 Programmer* (for creating and running programs) and *EV3 Commander* (for remote control) can also be used alongside the desktop software.

3.3.1 Brick Programming Language in EV3 Classroom

Block-based programming blocks (loops, switches, motor control, sensor reads) were used to structure behaviors quickly and visualize logic flows appropriate for demonstration and teaching.

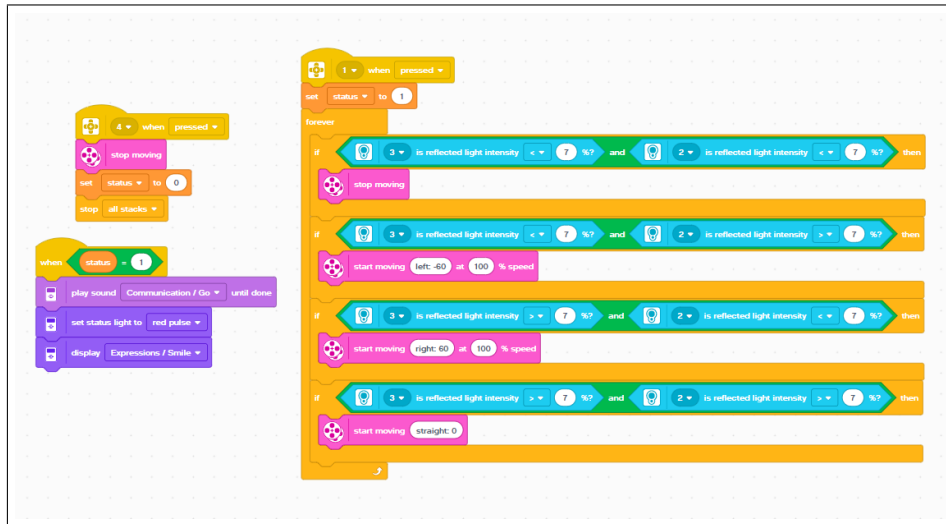


Figure 3.4: EV3 Classroom block program (e.g., line following).

3.3.2 Demonstrations of Control Logic

Representative behaviors developed and tested included:

- **Line following** using the color sensor to track a contrasting path.
- **Obstacle detection/avoidance** with the ultrasonic sensor to halt or reroute.
- **Pick-and-place** sequences on the Manipulator, coordinated with the conveyor start/stop.
- **Tool modularity** on TRACK3R to test actuation patterns with different end-effectors.

3.4 Hands-on Implementation

Hands-on sessions covered mechanical assembly, wiring/sensor placement, and program iteration. Emphasis was placed on stable gearing, proper belt tension (for the conveyor), and repeatable sensor positioning for consistent behavior across trials.

3.5 Challenges and Learnings

3.5.1 Challenges

- **Mechanical limits:** Load capacity and structural stiffness of brick builds constrained end-effector payloads and conveyor throughput.
- **Sensor variability:** Ambient light and surface reflectance affected color sensing; ultrasonic readings varied with target geometry.
- **Logic complexity:** Some state machines were cumbersome in pure block programming and required careful modularization.

3.5.2 Learnings

- Rapid prototyping through modular builds accelerated concept validation and teaching.
- Sensor-actuator integration practices (calibration, debouncing, thresholds) improved reliability.
- Clear interfaces between motion, sensing, and decision blocks simplified debugging and reuse.

Chapter 4

Workshop Sessions

4.1 Planning and Content of Sessions

The workshop sessions were structured to combine theoretical learning with practical exposure. They began with an **introduction to robotics**, covering essential concepts such as:

- Types of Robots: Industrial, service, mobile, humanoids
- Joints and Links: Mechanical building blocks of robots
- Types of Joints: Revolute, prismatic, cylindrical, spherical
- Workspaces: Reachable and dexterous workspaces
- Degrees of Freedom (DOF): Independent motions in robots
- Forward and Inverse Kinematics: Basics of robotic motion analysis

This foundation enabled participants to better understand subsequent demonstrations.

4.2 Demonstrations Using LEGO Kits

The **LEGO Mindstorms EV3 kits** were used to introduce robotics in an interactive manner. Students were shown models such as Everstorm, Raptor, Tracker, Manipulator, and a custom-built conveyor mechanism.

- Demonstrations included use of sensors, motors, and controllers.
- Students observed programming through the EV3 Brick software.
- Practical tasks showcased simple robotic functions.



Figure 4.1: Demonstration of LEGO EV3 Kits

4.3 Demonstrations of ABB IRB 120 Manipulator and Project Work

In addition to LEGO-based learning, students were introduced to **industrial robotics** through the **ABB IRB 120 manipulator** and its **IRC5 compact controller**. The session began with an overview of the manipulator's basic specifications, including

its six degrees of freedom (DOF), kinematic structure, and workspace characteristics. The functioning of the IRC5 compact controller was also explained to provide an understanding of how industrial robots are programmed and controlled.

Students were given the opportunity to **jog the robot joints** manually using the FlexPendant interface, which allowed them to explore robot motion in different joint configurations and Cartesian coordinates. Furthermore, the fundamentals of **RAPID programming language**, used for ABB robots, were introduced. Basic program structures, motion instructions, and execution methods were explained to familiarize students with industrial robot programming concepts.

The highlight of the demonstration was our **pick-and-place color sorting project**, developed using the ABB IRB 120. The workflow of integrating sensors, programming logic, and robot motion was explained in detail. Students observed how the manipulator could identify objects of different colors and perform automated pick-and-place operations.

Finally, comparisons were drawn between **educational kits (LEGO Mindstorms)** and **industrial manipulators (ABB IRB 120)**, emphasizing the transition from basic learning platforms to real-world industrial applications.

4.4 Hands-on Exposure Given to Students

Participants actively engaged in using the LEGO kits:

- Operating simple robots
- Programming movements and sensor responses
- Experimenting with basic robotic tasks

4.5 Feedback and Learning Outcomes

Feedback highlighted that participants:

- Gained better understanding of robotics fundamentals



Figure 4.2: Demonstration of ABB IRB 120 Manipulator and conveyor system

- Developed interest in practical robotic applications
- Became motivated for further research in robotics

The workshop successfully bridged theoretical knowledge with real-world applications.



Figure 4.3: Student Interaction During Workshop

Chapter 5

Universal Process Controller – Experimental Study

5.1 Introduction

The Universal Process Control Trainer (UPCT-330) from Apex Innovations is designed to study and analyze advanced process control techniques in a laboratory environment. It allows for the configuration of various control loops such as flow, level, pressure, and temperature, which are fundamental to the processing industries. The trainer integrates industrial-grade components with SCADA-based monitoring and a hybrid controller (Honeywell HC900), which makes it highly relevant for understanding automation practices in chemical, water treatment, and manufacturing systems. The system is also bundled with detailed hardware and software guides, as well as an experiment manual.

5.2 Experimental Setup

The setup consists of modules including a heating tank, centrifugal and plunger pumps, process tanks, pneumatic control valves with positioners, flow measuring devices (orifice, venturi, rotameter), differential pressure transmitters, I/P converters, solenoid valves, and a compressor. These are interconnected through piping and mounted on a standalone frame. The trainer interfaces with a computer via SCADA software, enabling loop configuration, data acquisition, and control tuning. Patch

board connections allow flexibility in selecting input and output variables for each experiment.

5.3 Experiments Performed (Part IV – Control Loops)

During the internship, Part IV of the experiment series was performed, focusing on closed-loop studies of flow, level, pressure, and temperature processes. These experiments provided practical understanding of process dynamics, controller action, and system limitations. A brief account of each experiment is given below.

IV-1: Flow Control Loop (Pump Outlet)

Objective: To study flow regulation at the pump outlet using a pneumatic control valve.

Summary: Flow processes are considered fast because changes in valve opening cause immediate changes in flow rate. The flow through a restriction can be approximated by:

$$Q = C_v \sqrt{\Delta P}$$

where Q is flow, ΔP is pressure drop, and C_v is the valve coefficient. The experiment showed quick response to setpoint changes with minor overshoot, confirming flow's fast dynamics and the need for proper valve calibration.

IV-2: Level Control Loop (Tank Inlet Control)

Objective: To maintain the water level in a tank by controlling the inlet valve.

Summary: Level is an integrating process, as inflow accumulates over time. The tank dynamics are given by:

$$A \frac{dh}{dt} = q_{in} - q_{out}$$

where A is cross-sectional area, h liquid height, and q_{in}, q_{out} flows. The system exhibited slow response, requiring integral action to eliminate steady-state offset, highlighting why level loops are slower than flow loops.



Figure 5.1: Univesal Process Control Trainer 330

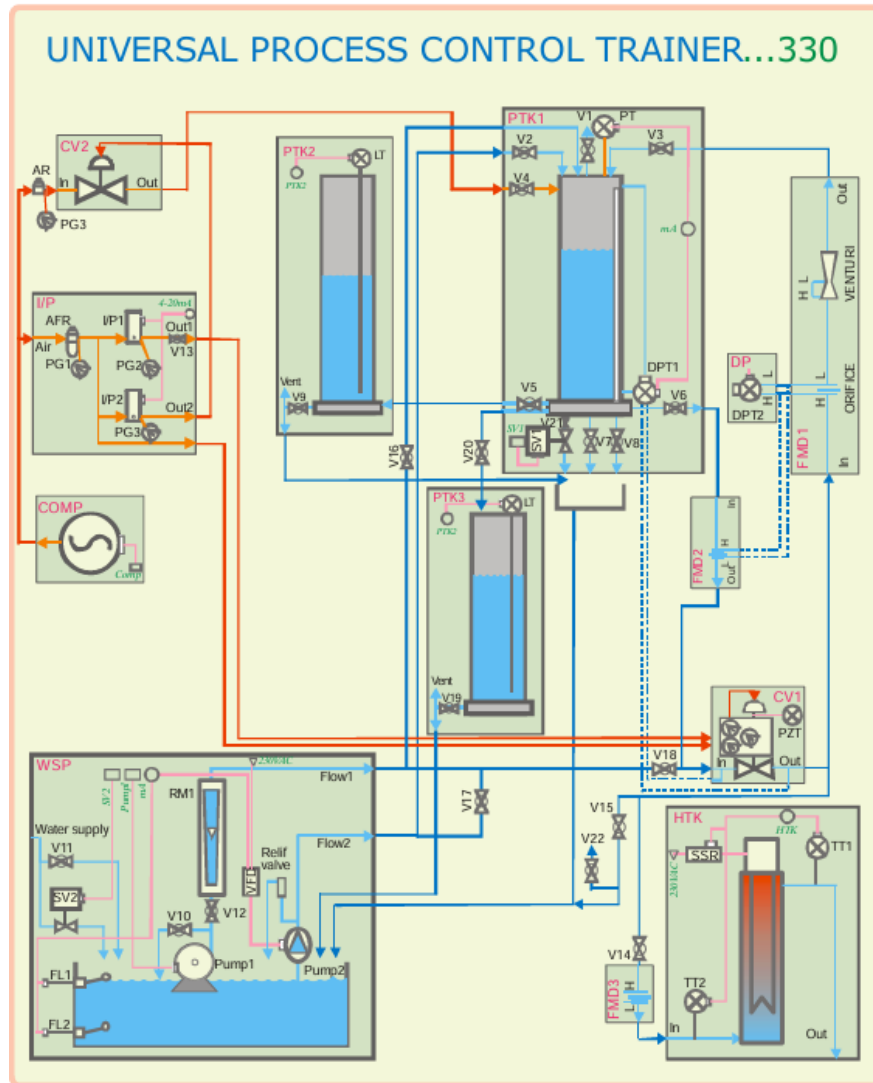


Figure 5.2: Schematic of the Universal Process Control Trainer (UPCT-330).

IV-3: Flow Control Loop (Tank Outlet)

Objective: To regulate tank outflow using a control valve at the outlet.

Summary: While outlet flow could be stabilized, the loop was sensitive to inflow variations. Outlet control lagged in disturbance rejection. The orifice flow relation is:

$$Q = C_d A \sqrt{2gH}$$

where C_d is discharge coefficient, A is orifice area, and H is liquid head. This demonstrated the importance of proper manipulated variable selection.

IV-4: Level Control Loop (Tank Outlet Control)

Objective: To control tank level by adjusting outlet valve opening.

Summary: The loop struggled with disturbance rejection, since inflow variations directly impacted tank level. Using the same tank equation as IV-2, outlet manipulation proved less effective for stable level control. This experiment reinforced the importance of aligning manipulated and disturbance variables.

IV-5: Level Control Loop (Plunger Pump)

Objective: To control tank level by regulating inflow with a positive displacement plunger pump.

Summary: The plunger pump provided precise inflow control compared to centrifugal pumps. Stroke adjustments directly influenced inflow, stabilizing the system once tuning accounted for slower level dynamics. This confirmed why displacement pumps are favored in dosing and precision applications.

IV-6: Pressure Control Loop (Water)

Objective: To regulate water pressure in the line.

Summary: Pressure is a moderately fast process variable. The relation is:

$$\Delta P = \frac{\rho Q^2}{2A^2}$$

where ρ is fluid density, Q flow, and A pipe area. Proportional-integral action ensured stable pressure control despite flow disturbances. The experiment emphasized the importance of transmitter calibration.

IV-7: Pressure Control Loop (Air)

Objective: To study pneumatic pressure regulation.

Summary: This experiment could not be conducted due to leakage in the water pump caused by gunk buildup. The failure highlighted the need for preventive maintenance in ensuring reliable loop operation and reflected real-world challenges

faced in process plants.

IV-8: Temperature Control Loop (Direct Heating)

Objective: To maintain water temperature in a heating tank using direct electric heating.

Summary: Temperature control is the slowest among process variables due to thermal lag. Heat balance is described by:

$$mC_p \frac{dT}{dt} = Q_{in} - Q_{loss}$$

where m is mass, C_p specific heat, and Q heat flows. The loop required careful tuning—aggressive gains caused oscillations while conservative settings gave sluggish response. This confirmed why temperature control is considered challenging.

5.4 Conclusion

The experiments provided hands-on exposure to fundamental process control loops and reinforced theoretical concepts. Key learnings included the difference in dynamics between flow, level, pressure, and temperature processes, the role of proper tuning in achieving stability, and the practical limitations posed by equipment conditions. This module successfully bridged theoretical control principles with real-world industrial practice.

Chapter 6

Overall Internship Outcomes

The internship provided structured exposure to industrial robotics, educational robotics, process control, and technical training activities. The outcomes achieved can be summarized based on the key chapters of the report.

Pick-and-Place Project Using ABB IRB 120

- Technical expertise in industrial robotic manipulator operation, including understanding DOF, workspace, and motion constraints.
- Hands-on experience with the IRC5 compact controller and RAPID programming for task-specific operations.
- Integration of sensor feedback (color sensor APDS-9960, limit switches) with robot motion sequences for deterministic pick-and-place functionality.
- System-level coordination of robot, conveyor (stepper motor-driven), and control logic for reliable operation.
- Practical skills in debugging, error handling, and optimizing end-effector performance for variable object sizes.

LEGO Mindstorms EV3 Robotics

- Enhanced understanding of modular robot design, sensor-actuator interfacing, and programming logic using LEGO EV3 kits.
- Implementation of autonomous robotic tasks using brick-based programming in EV3 Classroom.
- Experience in constructing predefined models (Everstorm, Raptor, Tracker, Manipulator) and a custom conveyor, providing insight into mechanical design considerations.
- Ability to translate theoretical robotic concepts into working prototypes for demonstration and testing.

Workshop Sessions

- Development of skills in planning and executing technical demonstrations for undergraduate participants.
- Technical ability to explain and illustrate fundamental robotics concepts: types of robots, joints and links, DOF, workspaces, forward and inverse kinematics.
- Experience in guiding students through LEGO EV3 kits and industrial robot demonstrations, including the pick-and-place project.
- Hands-on capability in managing interactive sessions, collecting feedback, and refining demonstration methodology for clarity and effectiveness.

Universal Process Control

- Practical understanding of industrial process control loops: flow, level, pressure, and temperature.
- Competence in configuring and analyzing control loops using the UPCT-330 trainer.

- Ability to perform PID tuning, interpret process dynamics, and assess system response under varying conditions.
- Exposure to device-level calibration, preventive maintenance considerations, and troubleshooting of industrial control equipment.
- Understanding of process optimization and the role of precise measurement and control in industrial automation.

Chapter 7

Conclusion and Future Scope

Conclusion

The internship at the Bosch Rexroth-CET Center of Excellence in Automation Technologies provided extensive exposure to both industrial and educational robotics, as well as process automation systems. Key conclusions from the internship are:

- **Industrial Robotics Competency:** Hands-on experience with the ABB IRB 120 manipulator and IRC5 compact controller reinforced understanding of robot kinematics, programming in RAPID, and system integration with external sensors and conveyors. Implementation of the pick-and-place color sorting project enhanced skills in end-effector selection, sensor interfacing, and synchronization between robot and peripheral systems.
- **Educational Robotics Expertise:** LEGO Mindstorms EV3 kits facilitated rapid prototyping, modular design, and programming logic implementation using brick-based language. Assembly and programming of multiple models (Everstorm, Raptor, Tracker, Manipulator, and custom conveyors) strengthened problem-solving, mechanical design, and control logic skills at a micro-scale.
- **Technical Communication and Training:** Planning and conducting workshop sessions for undergraduate students improved the ability to explain robotics concepts, demonstrate working models, and provide hands-on guidance in a structured manner. The experience highlighted the importance of pedagogical

methods in technical demonstrations and the ability to convey complex concepts in a simplified form.

- **Process Control Proficiency:** Study of the Universal Process Control Trainer (UPCT-330) deepened understanding of industrial process loops (flow, level, pressure, temperature), monitoring, PID tuning, and practical challenges in process automation. Exposure to system calibration, diagnostics, and preventive maintenance emphasized reliability and safety considerations in industrial setups.

Overall, the internship provided a holistic understanding of automation and robotics, combining industrial applications, educational tools, and process control theory into an integrated technical experience.

Future Scope

The internship experience opens multiple avenues for further development and enhancement:

- **Advanced Industrial Robotics:** Integration of vision-based systems (e.g., camera with ArUco markers, OpenCV) to reduce dependency on point sensors and enhance accuracy in pick-and-place operations. Exploration of more sophisticated end-effectors, adaptive grippers, and collaborative robot safety protocols.
- **Robotics Simulation and Programming:** Expansion of simulation-based testing in RobotStudio for offline programming and optimization of complex robotic tasks. Implementation of industrial-standard communication protocols (Ethernet/IP, PROFINET) for multi-robot and peripheral integration.
- **Educational Robotics Development:** Creation of advanced LEGO EV3 modules or custom kits for workshops to demonstrate concepts like automation sequencing, PID-based control, and sensor fusion. Development of structured learning modules for undergraduate students combining hands-on and simulation-based learning.

- **Process Automation and Control:** Application of learned control strategies to real-world industrial processes, including advanced process loops and multivariable control systems. Implementation of predictive maintenance, data logging, and performance optimization strategies in laboratory or industrial setups.
- **Research and Innovation Opportunities:** Integration of robotics and process control knowledge to develop smart automated systems for industrial and educational applications.

The internship has provided a robust foundation for pursuing advanced studies or professional work in robotics, automation, and process control, with a clear understanding of both theoretical principles and practical challenges.

References

- [1] ABB Robotics, “ABB IRB 120 Industrial Robot Product Manual,” ABB, 2020. [Online]. Available: <https://new.abb.com/products/robotics/industrial-robots/irb-120>
- [2] Arduino, “Arduino Uno and Mega Microcontrollers,” [Online]. Available: <https://www.arduino.cc>
- [3] LEGO Education, “LEGO MINDSTORMS EV3,” [Online]. Available: <https://education.lego.com/en-us/product/mindstorms-ev3>
- [4] APDS-9960 Datasheet, “Digital Proximity, Ambient Light, RGB, and Gesture Sensor,” Broadcom Inc., 2020.
- [5] LM393 Datasheet, “Low Power Dual Comparator,” Texas Instruments, 2020.
- [6] R. Siegwart, I. R. Nourbakhsh, and D. Scaramuzza, *Introduction to Autonomous Mobile Robots*, 2nd ed., MIT Press, 2011.